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Cross-Seasonal Optimal Dispatch of a Wind–Photovoltaic Hybrid Pumped-Storage Multi-Energy Complementary System

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Abstract

A hybrid pumped-storage power station (hereinafter referred to as the "hybrid station") is developed by installing reversible pump–turbine units between cascade reservoirs, providing both conventional hydropower regulation and cross-seasonal energy storage capability. However, existing studies have mainly focused on its short-term peak-shaving function, while the potential of cross-seasonal water resource allocation remains insufficiently explored. This paper develops a medium-to-long-term cross-seasonal optimal dispatch model considering the seasonal regulation characteristics of hybrid pumped-storage stations, with the objective of minimizing residual-load variance. Seasonal reservoir storage rate-of-change constraints are introduced to represent the operation mechanism of storing water during wet seasons and releasing water during dry seasons. A Variable-Phase Progressive Optimality Algorithm (VPPOA) is proposed to solve the model. Three operation modes, including no hybrid storage, hybrid daily regulation, and hybrid seasonal regulation, are established for comprehensive evaluation under different hydrological conditions, renewable outputs, load variations, and algorithm performances. The results show that the proposed model achieves an annual cross-seasonal energy transfer of 848.8 GWh. Compared with the no-hybrid-storage scheme, the hybrid seasonal regulation scheme reduces the residual-load variance by 77.3%, increases the wind–PV accommodation rate from 40.3% to 96.6%, and decreases the wet-season wind–PV curtailment rate from 38.8% to 3.9%. Compared with the hybrid daily regulation scheme, it further reduces the residual-load variance by 26.1%, improves the wind–PV accommodation rate by 9.2 percentage points, and increases dry-season supply reliability from 93.2% to 100%. Moreover, VPPOA outperforms GA, PSO, and conventional POA, achieving a lower residual-load variance of 2861.05 MW^2. The results demonstrate that the cross-seasonal regulation capability of hybrid pumped-storage stations can effectively enhance renewable energy accommodation and power supply reliability, providing support for the flexible operation of new-type power systems.

Keywords: wind–PV hybrid storage complementary system; seasonal regulation; cross-seasonal energy transfer; wind–PV accommodation; Variable-Phase Progressive Optimality Algorithm

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1. Introduction

With the proposal of carbon-peaking and carbon-neutrality goals, China's energy system is accelerating its transition toward a new-type power system dominated by wind

and photovoltaic (PV) power [1]. Meanwhile, emerging energy carriers and multi-energy integration pathways, including hydrogen-based energy systems, have attracted increasing attention as complementary approaches for large-scale renewable energy integration and energy transportation [2]. However, the strong volatility and randomness of renewable energy outputs increase power-balancing difficulties and impose higher flexibility requirements on power systems [3,4]. Especially under high renewable penetration, the spatiotemporal mismatch within the power system becomes more pronounced, and large-scale wind and PV curtailment occurs from time to time, weakening renewable accommodation capability and constraining further development [5,6]. Therefore, improving renewable energy accommodation capability through coordinated development of flexible resources, energy storage, and multi-energy complementary systems has become an essential pathway for achieving high-level renewable energy utilization. Against this backdrop, building power sources with strong regulation capability has become a key pathway for developing the new-type power system. Among these, hydropower and pumped-storage hydropower combine both generation and storage attributes, occupying an important strategic position in smoothing wind–PV fluctuations and enhancing system flexibility [7,8]. Meanwhile, cascade hydropower can effectively promote renewable accommodation through reservoir regulation [9]; however, hydropower alone or conventional pumped storage alone still struggles to fully cope with the impact of high-proportion wind–PV integration. Hydropower output during the dry season is constrained by seasonal runoff fluctuations, sharply reducing regulation capability; pumped storage, as a typical bidirectional "electricity–water energy" coupled system, exhibits significant fluctuation in its regulation capability with seasonal alternation and reservoir storage rise and fall, and its regulation value varies markedly under different operating strategies [10]. For this reason, building a multi-energy complementary system based on hydropower, led by wind–PV, and centered on pumped storage as the regulation core has become a current research hotspot. Meanwhile, the development of smart grid technologies and advanced communication infrastructures has provided technical support for real-time monitoring, information exchange, and coordinated operation of modern power systems, which is beneficial for improving the flexibility and intelligent management capability of renewable energy integration systems [11,12]. Among these, a hybrid pumped-storage power station (hereinafter referred to as a "hybrid station"), which combines both conventional hydropower and pumped-storage functions, demonstrates greater regulation flexibility and accommodation capability in cascade-reservoir regions. Studies show that by releasing long-duration regulation capability and shaving peaks/filling valleys, hybrid storage can significantly improve renewable utilization, providing effective support for grid integration of high-proportion renewable energy.

Existing studies have investigated the operational characteristics and optimal dispatch of hybrid pumped-storage stations from the perspectives of medium-to-long-term operation optimization, scheduling strategies, and operation rule formulation. Ai et al. [13] studied the medium-to-long-term compensatory peak regulation scheduling of cascaded hydropower stations under high renewable energy penetration, highlighting the importance of flexible hydropower regulation. Li et al. [14] developed a medium-to-long-term optimal dispatch model for cascade reservoirs with hybrid pumped-storage stations, demonstrating the contribution of pumping operation to increasing generation and firm output. Tang et al. [15] established a joint optimal dispatch model considering multiple operational objectives, including energy production, peak shaving, bidding, and energy storage. Cheng et al. [16] further proposed an operation rule modeling method for cascade hydropower systems with hybrid pumped-storage stations, providing a foundation for coordinated long-term operation. For short-term and multi-time-scale scheduling, existing

studies have mainly focused on peak shaving, renewable accommodation, and refined operation strategies. Luo et al. [17] developed a short-term optimal dispatch model for cascade hydro–photovoltaic complementary systems to maximize renewable energy accommodation capability. Cao et al. [18] established a refined operating-cost model based on flow optimization and analyzed the economic benefits of hybrid pumped-storage units under different operating conditions. Liu et al. [19] developed a short-term peak-shaving optimization model for cascade hybrid pumped-storage systems and effectively reduced the residual-load peak–valley difference through linearization methods. Qian et al. [20] proposed a short-term scheduling strategy for pump-station-integrated cascade hybrid pumped-storage systems, improving the regulation capability of multi-stage hybrid storage systems.

As the installed capacity of renewable energy continues to expand, hybrid pumped-storage stations have attracted increasing attention due to their advantages in smoothing wind–PV fluctuations, enhancing system flexibility, and improving renewable energy accommodation. Existing studies have mainly focused on long-cycle complementary operation, configuration optimization, short-term dispatch, and source–load coordination strategies. For long-cycle operation and configuration optimization, Zhang et al. [21] established a complementary generation optimal dispatch model for the Pankou-Xiaoxuan hybrid station, with the objective of minimizing residual load variance, and verified the improvement effect of hybrid storage on wind–PV accommodation. Peng et al. [22] further compared hybrid storage and pure pumped-storage configurations from the perspectives of load-tracking deviation, energy curtailment, and water utilization, demonstrating the advantages of hybrid storage in system coordination capability. For short-term operation and source–load matching, Feng et al. [23] developed an intraday complementary operation model for a hydro–PV storage system with hybrid pumped storage, considering output fluctuations of photovoltaic generation. Ji et al. [24] proposed a source–load-matching operation strategy for hydro–PV hybrid storage systems by optimizing pumping and generating periods, achieving improved coordination between renewable output and load demand. The existing studies mainly focus on hybrid pumped-storage operation optimization, short-term scheduling, renewable-storage coordination, and optimization algorithms. Table 1 summarizes the research progress, main methods, contributions, and limitations of existing studies on hydro–wind–PV storage complementary systems.

In summary, existing research has made important progress in the operating characteristics of hybrid stations, cascade hydropower dispatch methods, and hydro–wind–PV storage multi-energy complementary coordination. However, most existing studies focus on short-term fluctuation smoothing or specific scheduling scenarios [25–27], while insufficient attention has been paid to the cross-seasonal mismatch between renewable energy output and reservoir inflow. For hybrid pumped-storage stations with strong seasonal regulation capability, the potential of reallocating water resources between wet and dry periods has not been fully explored. Therefore, developing a cross-seasonal optimal dispatch model that considers the seasonal regulation characteristics of hybrid stations remains an important research problem. In addition, solving cross-seasonal hydro–wind–PV storage optimization problems remains challenging due to their nonlinear characteristics, multi-stage coupling relationships, and large-scale decision variables. Previous studies have applied various optimization algorithms, including mathematical programming methods and metaheuristic approaches, to renewable energy scheduling problems [28–30]. Although metaheuristic algorithms have demonstrated advantages in solving complex optimization problems, they usually suffer from slow convergence and difficulties in balancing exploration and exploitation when applied to large-scale nonlinear scheduling problems. Nevertheless, traditional progressive optimality algorithms generally adopt fixed opti-

mization strategies, which may limit their efficiency in solving complex multi-time-scale dispatch problems. Therefore, this study develops a Variable-Phase Progressive Optimality Algorithm (VPPOA) to improve convergence efficiency, optimization stability, and solution performance. Different from conventional progressive optimality algorithms with fixed iteration strategies, the proposed VPPOA introduces a variable-phase optimization mechanism that dynamically adjusts the optimization process according to the convergence characteristics of the solution. By dividing the optimization procedure into different phases with adaptive search strategies, VPPOA balances global exploration and local refinement, thereby improving convergence efficiency and solution stability for complex multi-time-scale hydro–wind–PV storage scheduling problems.

Table 1. Summary of existing studies on hydro–wind–PV storage complementary systems.

Research Category	Representative References	Main Methods	Main Contributions	Limitations
Medium- and long-term operation	[14–17]	Dynamic optimization; operation rule	Improved regulation capability of hybrid pumped-storage systems	Seasonal water–energy coordination insufficient
Short-term scheduling	[18–20]	Mathematical programming; optimization methods	Improved peak shaving and renewable accommodation	Lack of cross-seasonal reservoir coordination
Hydro–wind–PV storage coordination	[21–24]	Multi-objective optimization; source–load matching	Improved coordination among renewable energy, storage, and load	Seasonal regulation mechanism not fully considered
Optimization methods	[25–30]	Genetic algorithm; heuristic algorithms; and improved optimization methods	Improved computational efficiency	Difficult to solve large-scale nonlinear coupled problems
Seasonal regulation of hybrid pumped storage	[10,31]	Seasonal operation optimization	Revealed the potential of seasonal storage regulation	Cross-seasonal optimal dispatch models require further development

In view of this, the main contributions of this study are summarized as follows:

(1) A medium-to-long-term cross-seasonal optimal dispatch model for a hydro–wind–PV storage complementary system with a hybrid pumped-storage station is established, where seasonal reservoir storage rate-of-change constraints are introduced to characterize wet-season storage and dry-season release processes.
(2) A Variable-Phase Progressive Optimality Algorithm (VPPOA) is developed by introducing a phase-adaptive optimization mechanism, which improves convergence efficiency and solution stability for complex multi-time-scale scheduling problems.
(3) A comprehensive evaluation framework considering operation strategies, hydrological conditions, renewable output levels, load variations, and algorithm comparisons is constructed to verify the effectiveness of the proposed model.

2. Materials and Methods

2.1. Cross-Seasonal Optimal Dispatch Model for a Wind–PV Hybrid Storage Multi-Energy Complementary System

2.1.1. Objective Function

$$f = \min \sum_{t=1}^{T} \left[R_t - \frac{1}{T} \sum_{t=1}^{T} R_t \right]^2 \quad (1)$$

$$R_t = P_t^{net} - P_t^h + P_t^p \quad (2)$$

$$P_t^{net} = P_t - P_t^w - P_t^{pv} \tag{3}$$

where f is the minimum residual-load variance; R_t is the residual load; T is the total number of dispatch periods; P_t^{net} is the net load of the wind–PV hybrid system in period t; P_t^h and P_t^p are, respectively, the generating output and pumping power of the hybrid station in period t; P_t is the system load in period t; and P_t^w and P_t^{pv} are, respectively, the total wind power output and total PV output in period t.

2.1.2. Constraints

(1) Water balance constraints:

$$V_{1,t+1} = V_{1,t} + \left(I_{1,t} - q_{1,t} + Q_{p,t}\right) \cdot \Delta t \tag{4}$$

$$V_{2,t+1} = V_{2,t} + \left(I_{2,t} + q_{1,t} - q_{2,t} - Q_{p,t}\right) \cdot \Delta t \tag{5}$$

where $V_{1,t}$ and $V_{1,t+1}$ are, respectively, the storage volumes of the upper reservoir at the start of period t and period $t+1$; $I_{1,t}$ and $q_{1,t}$ are, respectively, the average inflow and average outflow of the upper reservoir in period t; $V_{2,t}$ and $V_{2,t+1}$ are, respectively, the storage volumes of the lower reservoir at the start of period t and period $t+1$; $I_{2,t}$ and $q_{2,t}$ are, respectively, the average inflow and average outflow of the lower reservoir in period t; $Q_{p,t}$ is the pumping flow of the hybrid station in period t; and Δt is the length of a time period.

(2) Reservoir storage capacity constraints:

$$V_{1,t}^{\min} \le V_{1,t} \le V_{1,t}^{\max} \tag{6}$$

$$V_{2,t}^{\min} \le V_{2,t} \le V_{2,t}^{\max} \tag{7}$$

where $V_{1,t}^{\min}$ and $V_{1,t}^{\max}$ are, respectively, the dead storage capacity and the maximum allowable storage capacity of the upper reservoir in period t, corresponding to the flood-control limit water level during the flood season and to the normal pool level during other periods; $V_{2,t}^{\min}$ and $V_{2,t}^{\max}$ are, respectively, the dead storage capacity and the maximum allowable storage capacity of the lower reservoir in period t.

(3) Seasonal constraint on the pumped-storage reservoir [31]:

$$V_{1,t} - V_{1,t-1} \begin{cases} \ge \Delta V_s^{in}, s \in \Omega^{flood} \\ \le -\Delta V_s^{out}, s \in \Omega^{dry} \\ \in [-\varepsilon, +\varepsilon], s \in \Omega^{trans} \end{cases} \tag{8}$$

where $V_{1,t-1}$ is the storage capacity of the upper reservoir in period $t-1$; ΔV_s^{in} and ΔV_s^{out} are, respectively, the minimum storage-capacity increment during storing phase s and the minimum storage-capacity decrement during releasing phase s; ω is the maximum allowable storage-capacity fluctuation during the transition phase; and Ω^{flood}, Ω^{dry}, and Ω^{trans} are, respectively, the sets of time periods for the flood season, the dry season, and the transition season. In this paper, the flood season corresponds to May–September, the dry season to December–February of the following year, and the transition season to March–April and October–November.

(4) Release-flow constraints:

$$Q_{1,\min} \le q_{1,t} \le Q_{1,\max} \tag{9}$$

$$Q_{2,\min} \le q_{2,t} \le Q_{2,\max} \tag{10}$$

where $Q_{1,\min}$ and $Q_{1,\max}$ are, respectively, the minimum ecological release flow and the maximum release flow of the upper reservoir; $Q_{2,\min}$ and $Q_{2,\max}$ are, respectively, the minimum ecological release flow and the maximum release flow of the lower reservoir.

(5) Hydropower output constraints:

$$N_{1,\min} \le P_{h,1,t} \le N_{1,\max} \tag{11}$$

$$N_{2,\min} \le P_{h,2,t} \le N_{2,\max} \tag{12}$$

where $P_{h,1,t}$ and $P_{h,2,t}$ are, respectively, the total generating output of the upper- and lower-reservoir hydropower stations in period t; $N_{1,\min}$ and $N_{1,\max}$ are, respectively, the minimum and maximum generating output of the upper-reservoir hydropower station; and $N_{2,\min}$ and $N_{2,\max}$ are, respectively, the minimum and maximum generating output of the lower-reservoir hydropower station.

(6) Decomposition of the upper-reservoir hydropower station's total output:

$$P_{h,1,t} = P_{c,1,t} + P_t^h \tag{13}$$

where $P_{c,1,t}$ is the generating output of the upper-reservoir hydropower station's conventional units in period t.

(7) Conventional-unit output constraint:

$$N_{c,1,\min} \le P_{c,1,t} \le N_{c,1,\max} \tag{14}$$

(8) Reversible-unit output constraints:

In generating mode:

$$P_{h,\min} \le P_{h,t} \le P_{h,\max} \tag{15}$$

In pumping mode:

$$P_{p,\min} \le P_{p,t} \le P_{p,\max} \tag{16}$$

where $P_{h,\min}$ and $P_{h,\max}$ are, respectively, the minimum and maximum power of the reversible unit in generating mode; $P_{p,\min}$ and $P_{p,\max}$ are, respectively, the minimum and maximum power of the reversible unit in pumping mode.

(9) Reversible-unit flow constraint:

$$Q_{p,\min} \le Q_{p,t} \le Q_{p,\max} \tag{17}$$

where $Q_{p,t}$ is the pumping flow of the pumped-storage unit in period t; $Q_{p,\min}$ and $Q_{p,\max}$ are, respectively, the minimum and maximum pumping flow of the hybrid unit.

(10) Relationship between pumping flow and power:

$$P_t^p = \frac{\rho g H_t Q_{p,t}}{1000 \eta_p} \tag{18}$$

where ρ is the density of water; g is the acceleration of gravity; H_t is the pumping head; and η_p is the pumping efficiency.

(11) Mutual-exclusivity constraint between pumping and generating modes:

$$N_{g,t} \cdot N_{p,t} = 0 \tag{19}$$

This constraint indicates that the reversible unit cannot be in both generating and pumping modes within the same period.

2.1.3. Dispatch Performance Evaluation Indicators

To comprehensively evaluate the dispatch performance of the proposed wind–PV hybrid storage system, several indicators are introduced from the perspectives of renewable energy utilization, output stability, power supply reliability, and cross-seasonal energy regulation capability. The calculation methods of the key performance indicators are defined as follows.

(1) Wind–PV curtailment:

$$E_{cur} = \sum_{t=1}^{T} \max(P_{RE,t} - P_{acc,t}, 0) \cdot \Delta t \tag{20}$$

where E_{cur} represents the total wind–PV curtailment energy; $P_{RE,t}$ denotes the available renewable energy output at time; and $P_{acc,t}$ represents the accommodated renewable energy power.

(2) Wind–PV accommodation rate:

$$\eta_{RE} = \left(1 - \frac{E_{cur}}{E_{RE}}\right) \times 100\% \tag{21}$$

where η_{RE} is the wind–PV accommodation rate, and E_{RE} denotes the total available wind–PV generation energy.

(3) Dry-season supply reliability:

$$R_{dry} = \left(1 - \frac{E_{loss,dry}}{E_{load,dry}}\right) \times 100\% \tag{22}$$

where R_{dry} represents the dry-season supply reliability; $E_{loss,dry}$ denotes the electricity supply deficit during the dry season; and $E_{load,dry}$ represents the total load demand during the dry season.

(4) Wet-season wind–PV curtailment rate:

$$C_{wet} = \frac{E_{cur,wet}}{E_{RE,wet}} \times 100\% \tag{23}$$

where C_{wet} represents the wet-season wind–PV curtailment rate; $E_{cur,wet}$ denotes the wind–PV curtailment energy during the wet season; and $E_{RE,wet}$ represents the available wind–PV generation energy during the wet season.

(5) Cross-seasonal transferred energy:

$$E_{cs} = \sum_{t=1}^{T} P_{trans,t} \cdot \Delta t \tag{24}$$

where E_{cs} represents the energy transferred across seasons; $P_{trans,t}$ denotes the cross-seasonal energy transfer power at time t.

2.2. Uncertainty Scenario Generation

Wind and PV outputs, together with system load, exhibit significant random fluctuations, while inflow shows pronounced seasonal wet–dry variations and substantial inter-annual differences. To further evaluate the adaptability of the proposed cross-seasonal optimal dispatch method under different operating conditions, uncertainty scenario analysis is incorporated into the original dispatch model. Specifically, the Latin Hypercube Sampling (LHS) method combined with K-means clustering is employed to generate representative stochastic scenarios for wind power, PV output, and system load, while

three hydrological scenarios, representing wet, normal, and dry years, are constructed using the typical hydrological year method.

2.2.1. Stochastic Scenario Generation for Wind Power, PV Output, and System Load

The historical observations of wind power, PV output, and system load are denoted by $P_{w,t}$, $P_{pv,t}$ and $P_{L,t}$, respectively, which together form the random vector:

$$X_t = \left[P_{w,t}, P_{pv,t}, P_{L,t}\right] \tag{25}$$

where $t = 1, 2, \cdots, T$ denotes the dispatch period.

(1) Generation of initial random scenarios using Latin Hypercube Sampling (LHS): The probability interval $[0, 1]$ is equally divided into N_0 subintervals, and one sample is randomly selected from the k-th subinterval:

$$U_k = \frac{n - 1 + \xi_k}{N_0}, \xi_k \sim (0, 1), k = 1, 2, \cdots, N_0 \tag{26}$$

(2) Inverse transformation based on empirical cumulative distribution functions: The corresponding random samples of wind power, PV output, and system load are obtained by applying the inverse transformation to their respective empirical cumulative distribution functions:

$$P_{w,k} = F_W^{-1}\left(u_k^W\right), P_{pv,k} = F_{PV}^{-1}\left(u_k^{PV}\right), P_{L,k} = F_L^{-1}\left(u_k^L\right) \tag{27}$$

where F_W, F_{PV} and F_L denote the empirical cumulative distribution functions of wind power output, PV output, and system load, respectively; N_0 denotes the number of initial random scenarios.

(3) Scenario reduction using K-means clustering: After a large number of random scenarios are generated using the above procedure, K-means clustering is employed to reduce the scenario set. The clustering objective is formulated as

$$\min \sum_{s=1}^{N_s} \sum_{i \in C_s} \|X_i - \mu_s\|_2^2 \tag{28}$$

where N_s denotes the number of representative scenarios after reduction, C_s denotes the s-th cluster, and μ_s denotes the corresponding cluster centroid.

(4) Probability assignment for representative scenarios. The representative wind–PV–load scenarios are obtained after scenario reduction, and the probability of each representative scenario is determined by the proportion of original scenarios assigned to the corresponding cluster:

$$\pi_s = \frac{n_s}{N_0}, s = 1, 2, \cdots, N_s \tag{29}$$

where n_s denotes the number of original scenarios contained in the s-th cluster. Since N_0 all original scenarios are assigned to one of the N_s clusters, the scenario probabilities satisfy:

$$\sum_{s=1}^{N_s} \pi_s = 1 \tag{30}$$

2.2.2. Hydrological Uncertainty Scenarios

Given the pronounced seasonal wet–dry variations and interannual differences in reservoir inflow, the typical hydrological year method is adopted to construct three hydro-

logical scenarios, namely, wet, normal, and dry years. The total annual inflow for the -th year is calculated as

$$W_y = \sum_{t=1}^{T} Q_{y,t} \Delta t \tag{31}$$

where $Q_{y,t}$ denotes the inflow rate during the t-th period of the y-th year, T is the total number of periods in a year, and Δt denotes the length of each period.

The annual total inflow values of all historical years are ranked, and representative wet, normal, and dry years are selected accordingly. The corresponding annual inflow processes are then adopted as the typical hydrological scenarios.

2.2.3. Combined Uncertainty Scenarios

After the LHS and K-means procedures, $N_R = 5$ representative wind–PV–load scenarios are obtained. Combined with the three hydrological scenarios, the total number of combined uncertainty scenarios is 15. The -th combined scenario is expressed as

$$\Omega_s = \left\{ P_{w,t}^{(s)}, P_{pv,t}^{(s)}, P_{L,t}^{(s)}, Q_t^{(s)} \right\}_{t=1}^{T} \tag{32}$$

where $P_{w,t}^{(s)}$, $P_{pv,t}^{(s)}$, $P_{L,t}^{(s)}$ and $Q_t^{(s)}$ denote the wind power output, PV output, system load, and inflow, respectively, under the -th combined scenario.

Considering that a large number of similar scenarios may increase the computational burden of the optimization model, and that this study focuses on the cross-seasonal regulation capability of the wind–PV hybrid pumped-storage multi-energy complementary system under both typical and adverse operating conditions, six representative scenarios are selected from the 15 combined scenarios for detailed analysis. The selected scenarios cover a range of operating conditions, including the baseline condition, low wind–PV output, high load, dry-year conditions, combined low wind–PV output and high load, and comprehensive adverse conditions. The specific scenario settings are presented in Table 2. The above typical combined uncertainty scenarios provide the inputs for subsequent optimal dispatch under different operating conditions.

Table 2. Representative scenarios for uncertainty analysis.

Scenario ID	Scenario Name	Wind–PV Output Level	Load Level	Hydrological Condition
S1	Baseline Scenario	Typical	Typical	Normal Year
S2	Low-Wind–PV Scenario	Low	Typical	Normal Year
S3	High-Load Scenario	Typical	High	Normal Year
S4	Dry-Year Scenario	Typical	Typical	Dry Year
S5	Low-Wind–PV + High-Load Scenario	Low	High	Normal Year
S6	Comprehensive Adverse Scenario	Low	High	Dry Year

2.3. Model Solution

For each typical scenario constructed in Section 2.2, the same VPPOA-based optimization procedure is independently applied to solve the cross-seasonal medium-to-long-term dispatch model for the wind–PV hybrid storage system. Through multiple initial trajectories, variable phase lengths, and recursive window search, the algorithm achieves fast convergence and global search capability for high-dimensional nonlinear dispatch problems. The model-solving procedure is shown in Figure 1, with the specific solution steps as follows:

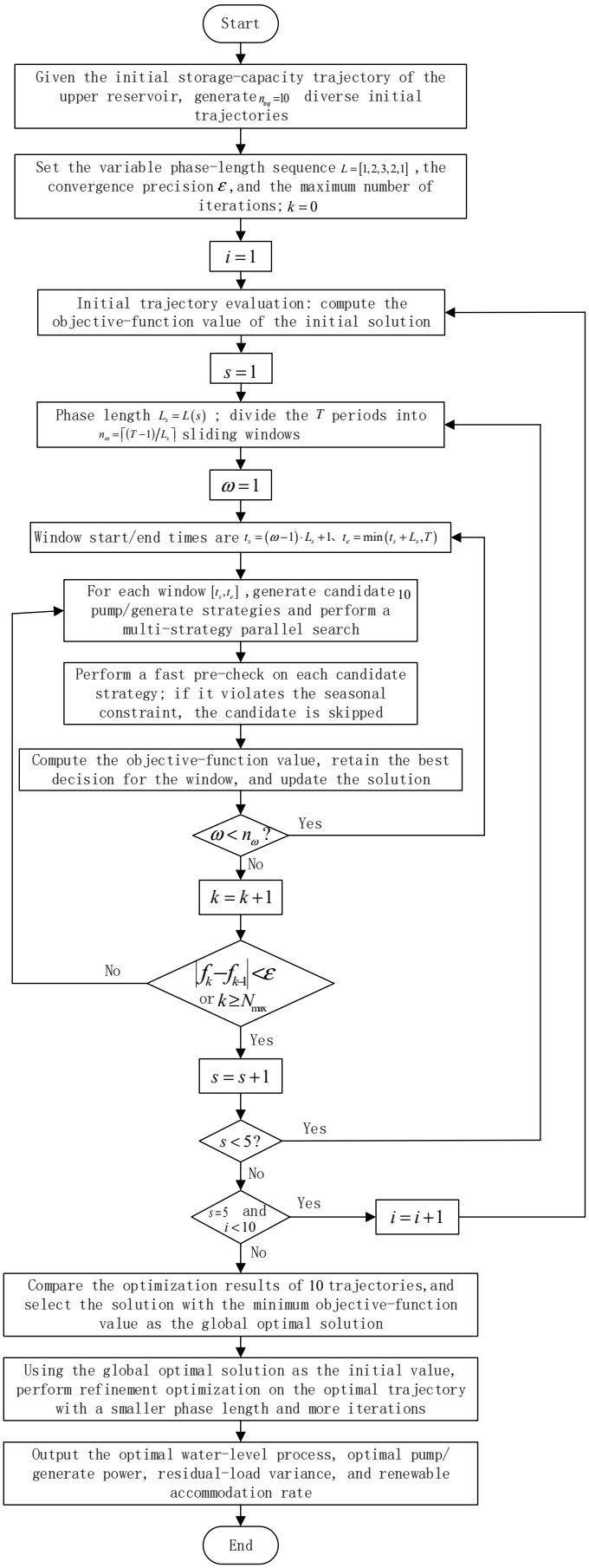


Figure 1. Flowchart of the Variable-Phase Progressive Optimality Algorithm (VPPOA).

Step 1: Initialization. Divide the annual dispatch period into T time periods, and determine the upper and lower storage capacity bounds $V_{1,t}^{\min}$ and $V_{1,t}^{\max}$ of the upper reservoir, together with the unit parameters $P_{h,\max}$ and $P_{p,\max}$; specify an initial storage capacity trajectory for the upper reservoir $\left\{V_{1,t}^{0}\right\}_{t=1}^{T+1}$, where $t = 1, 2, \cdots, T+1$ denotes the start of each period. To avoid local optima and enhance cross-seasonal regulation capability, generate $n_{traj} = 5$ diverse initial trajectories, each of which takes the form $X^{(k)} = \left\{V_1^{(k)}(t), P_h^{(k)}(t), P_p^{(k)}(t)\right\}, k = 1, \cdots, n_{traj}$. The initial trajectories are constructed using the following methods: a constant-water-level trajectory, a cross-seasonal heuristic trajectory, a reverse inflow-tracking trajectory, a load-tracking trajectory, and a wind–PV output-tracking trajectory. Run one fast annual simulation for each initial trajectory to obtain an initial objective function value as the starting point for subsequent optimization. To progressively approach the global optimum, set the variable phase-length sequence $L = [1, 2, 3, 2, 1]$, the convergence precision ε, and the maximum number of iterations $N_{\max}$; set the trajectory index $i = 1$, the phase index $s = 1$, and the iteration counter $k = 0$.

Step 2: Variable-Phase Progressive Optimization. Perform variable-phase optimization on the current trajectory i, with phase index $s = 1, 2, \cdots, 5$ and phase length $L_s = L(s)$. Divide the T periods into $n_\omega = \lceil (T-1)/L_s \rceil$ sliding windows, with window index $\omega = 1, 2, \cdots, n_\omega$, and window start/end times given by $t_s = (\omega - 1) \cdot L_s + 1$ and $t_e = \min(t_s + L_s, T)$. Boundary conditions are determined as follows: if $t_s = 1$, then $V_{1,s} = V_1^0$; otherwise, $V_{1,s} = V_1(t_s)$. If $t_e = T$, then $V_{1,e} = V_1^0$; otherwise, $V_{1,e} = V_1(t_e + 1)$. Then, for each window, perform the within-window candidate pump/generate strategy optimization (see Step 3). After each phase, recompute the objective function, and if the phase has already converged, break out early, recording the trajectory with the best objective value among all phases.

Step 3: Sliding-Window Optimization. For each window $[t_s, t_e]$, generate $n_{trial} = 10$ candidate pump/generate strategies $\left\{P_{h,t}^{try}, P_{p,t}^{try}\right\}_{t=t_s}^{t_e}$, and perform a multi-strategy parallel candidate search, including a load-tracking strategy, a strong pumping during high wind–PV output strategy, a forced full-generation at load peak strategy, a priority wind–PV accommodation strategy, an output-smoothing strategy, a rate-of-change-limiting strategy, and a random perturbation strategy. Each candidate strategy generates a new set of (P_h, P_p), and a fast water balance forecast is carried out via Equation (6); if a candidate violates the seasonal constraint, it is skipped. For candidates passing the pre-check, the annual simulation model is invoked to compute the objective function value, and the best candidate is selected to update the global trajectory.

Step 4: Iterative Convergence Check. Update the iteration counter $k = k + 1$, and compute the change in the objective value between adjacent iterations, $\Delta f = |f_k - f_{k-1}|$. If $\Delta f < \varepsilon$ or $k \geq N_{\max}$, the phase is deemed converged, and the algorithm proceeds to the next phase; otherwise, return to Step 3. After phase optimization is complete, if $s < 5$, set $s = s + 1$ and continue variable-phase optimization; if $s = 5$ and $i < n_{traj}$, set $i = i + 1$ and return to Step 3; otherwise, proceed to Step 5.

Step 5: Global Trajectory Competition. After all initial trajectories have completed phase optimization, compare the optimization results of all n_{traj} trajectories and select the solution with the minimum objective function value as the best solution obtained by VPPOA: $i^* = \text{argmin}_{i=1,\cdots,n_{traj}}\{f_i^*\}$, $\left\{V_1^*, P_h^*, P_p^*\right\} = \left\{V_{1,i}, P_{h,i}, P_{p,i}\right\}$, recording the optimal trajectory together with its source trajectory index and phase index.

Step 6: Final Refinement. Using the best solution obtained by VPPOA as the initial value, perform refinement optimization on the optimal trajectory with a smaller phase length and more iterations. Set the phase length $L = 1$, the maximum number of iterations

$2N_{\max}$, and the convergence precision $\varepsilon/10$, and carry out the refinement to ensure the final result achieves a fine-grained optimum within its local neighborhood.

Step 7: Output Results. Output the optimal water-level process $\left\{V_{1,t}^{*}\right\}_{t=1}^{T+1}$, the optimal pump/generate power $\left\{P_{h,t}^{*}, P_{p,t}^{*}\right\}_{t=1}^{T}$, the residual-load variance f^{*}, and the total curtailed energy $E_{curtail}^{*}$, together with the renewable accommodation rate $\eta^{*} = 1 - \frac{E_{curtail}^{*}}{\sum_{t=1}^{T}\left(P_{t}^{w}+P_{t}^{pv}\right)\cdot\Delta t}$, ultimately obtaining the annual cross-seasonal optimal dispatch scheme.

3. Results

3.1. Case Setup

This paper selects a wind–PV hybrid storage multi-energy complementary system composed of a cascade of hydropower stations in a certain river basin and the surrounding wind and PV resources as the research object; a schematic diagram is shown in Figure 2.

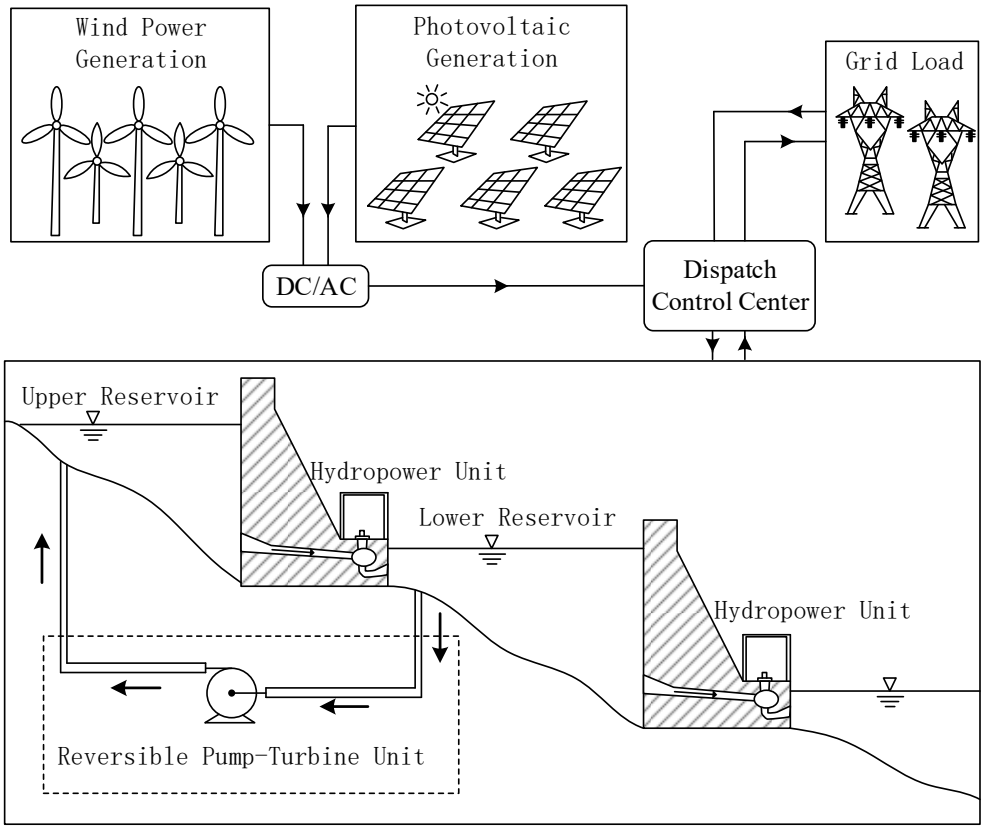


Figure 2. Schematic diagram of the wind–PV hybrid storage multi-energy complementary system.

The cascade consists of two hydropower stations; the basic characteristics of the two stations' reservoirs are shown in Table 3. Station A's reservoir is an annual regulation reservoir with a relatively large storage capacity and regulation capability, providing the natural conditions for cross-seasonal energy storage dispatch; reversible units are added on this basis to form the hybrid station. Station B's reservoir is a daily regulation reservoir and participates in system operation as a conventional hydropower station. According to the region's clean-energy development plan, the installed capacity ratio of hydropower, wind power, and PV is approximately 0.46:0.12:0.42. Taking the total installed capacity of 848 MW for the hybrid station after Station A's expansion as the hydropower benchmark, the wind farm and PV station are sized according to this ratio at 221 MW and 774 MW, respectively. Based on this system composition and installed capacity, the

dispatch period in this paper is set to one year, fully covering the inflow and wind–PV output characteristics of the flood, dry, and transition seasons, which can better reflect the seasonal differences in energy supply and demand over a long-cycle time scale. This paper uses meteorological and hydrological data for 2014, with the meteorological data sourced from the European Centre for Medium-Range Weather Forecasts (ECMWF). The annual inflow in 2014 is above the multi-year average, indicating that 2014 can be broadly regarded as a relatively wet hydrological year. This choice provides a representative and relatively demanding test case for evaluating the cross-seasonal regulation capability of the hybrid pumped-storage system, particularly its ability to accommodate seasonal surplus energy and mitigate renewable energy curtailment with relatively high water availability. The complete and continuous time series ensures consistent and reliable input conditions for the dispatch model.

Table 3. Basic characteristics of the hydropower stations' reservoirs.

Parameter	Upper Reservoir (A)	Lower Reservoir (B)
Total storage capacity (10^4 m^3)	233,800	3678
Normal pool level (m)	355	265
Dead water level (m)	330	260
Regulating storage capacity (10^4 m^3)	112,000	940
Regulation capability	Annual regulation	Daily regulation
Conventional-unit installed capacity (MW)	500	48
Reversible-unit installed capacity (MW)	149 × 2	—
Minimum ecological release flow (m^3/s)	32.8	16.7

3.2. Scenario Setup

To investigate the impact of the hybrid station and different dispatch modes on the wind–PV hybrid storage multi-energy complementary system, the seasonal distribution characteristics of the region's hydro–wind–PV resources are first analyzed; the intra-annual variation characteristics of Station A's inflow and wind–PV output are shown in Figure 3. As Figure 3 shows, during the dry season (January–February and December), inflow is at its lowest level of the year, while wind–PV output remains relatively high; during the transition season (March–April), inflow begins to recover, and wind–PV output fluctuates slightly but remains generally strong overall; during the wet season (May–September), inflow increases substantially and peaks around September, but the increase in wind–PV output over the same period is clearly smaller than the increase in runoff; and during the transition season (October–November) inflow recedes rapidly and wind–PV output declines accordingly. Overall, the 2014 data show a pronounced seasonal resource complementarity pattern in the study region: wet-season inflow is abundant while wind–PV output is relatively weak, whereas dry-season inflow is scarce, while wind–PV output is relatively strong. During the flood season, hydropower output is ample but the room for wind–PV accommodation is limited, while during the dry season, wind–PV output is relatively strong but hydropower regulation capability is insufficient, providing a natural resource complementarity basis for the hybrid station to achieve energy time shifting through cross-seasonal storage and release.

Based on the above resource characteristics, this paper sets up three comparison schemes—no hybrid storage, hybrid daily regulation, and hybrid seasonal regulation—to systematically analyze the improvement effects of hybrid station configuration and cross-seasonal dispatch strategy on output stability, renewable accommodation, and reservoir storage utilization. The specific setup of each scheme is as follows:

(1) No-hybrid-storage scheme: A conventional hydro–wind–PV complementary system consisting of the cascade hydropower stations together with the regional wind farm

and PV station. The hydropower stations are equipped only with conventional turbine-generator units, without pumped-storage units; the reservoirs are dispatched for generation following natural inflow within the storage-capacity bounds. This serves as the baseline scenario for quantifying the optimization space offered by the hybrid station and cross-seasonal dispatch.

(2) Hybrid daily regulation scheme: Building on the conventional hydro–wind–PV system, reversible units are added at Station A to form the hybrid station; at the medium-to-long-term scale, the hybrid station's seasonal regulation role is not considered, and the water pumped up within a day is released again within the same day. Therefore, the medium-to-long-term water-level process from the no-hybrid-storage scheme is taken as the water-level boundary for this scheme; within this water-level boundary, the dispatch model freely seeks the optimum only within the storage capacity and flow constraints, with the hybrid unit flexibly pumping/generating to accommodate curtailed wind–PV power and smooth system output fluctuations. This is used to quantify the regulation value of a stand-alone hybrid storage configuration at the intraday scale, providing a comparison baseline for evaluating the advantages of the cross-seasonal dispatch scheme.

(3) Hybrid seasonal regulation scheme: The system composition is identical to the hybrid daily regulation scheme, the difference being that the medium-to-long-term water-level boundary is obtained by introducing seasonal reservoir storage rate-of-change constraints, actively guiding the hybrid station's cross-seasonal storage/release behavior. During the flood season, storage capacity is required to increase continuously, driving the hybrid station to prioritize pumping and storing; during the dry season, storage capacity is required to decrease continuously, guiding the reservoir to increase release and generate using the stored water energy to make up load shortfalls; and during the transition season, storage capacity is allowed to fluctuate within a small range, balancing storage and generation needs to achieve cross-seasonal energy transfer from wet to dry periods.

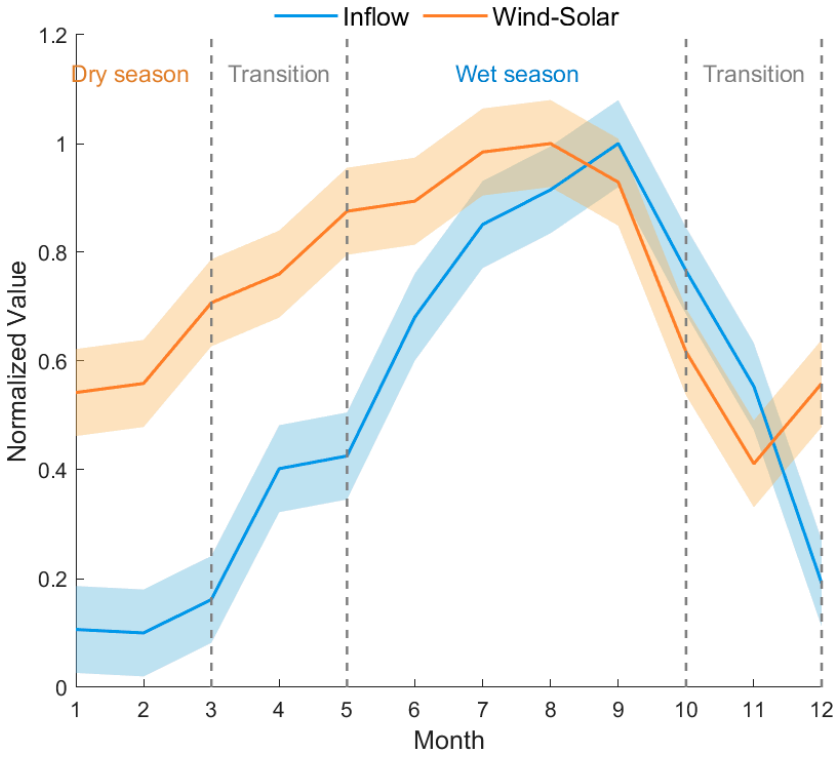


Figure 3. Intra-annual variation characteristics of Station A's inflow and wind–PV output.

3.3. Result Analysis

3.3.1. Output Process and Water-Level Analysis

The output processes of the wind–PV hybrid system under the three schemes are shown in Figure 4. As Figure 4 shows, in Scheme 1, hydropower generates continuously throughout the year but with limited regulation capability; during the flood season, combined with large PV output, total output significantly exceeds load, causing curtailment, while during the dry season, insufficient inflow makes it difficult to meet supply demand. Scheme 2 achieves relatively good tracking of load by total output through pumped storage regulation, converting surplus electricity into potential energy during the flood season, but because energy cannot be transferred between seasons, hydropower output during the dry season still remains at a relatively low level. Scheme 3, aiming to achieve storage in wet periods to supplement dry periods, substantially reduces hydropower output during the flood season while maintaining a high reservoir level, and significantly increases hydropower output during the dry season, successfully transferring surplus flood-season energy for use in the dry season and markedly improving dry-season supply reliability.

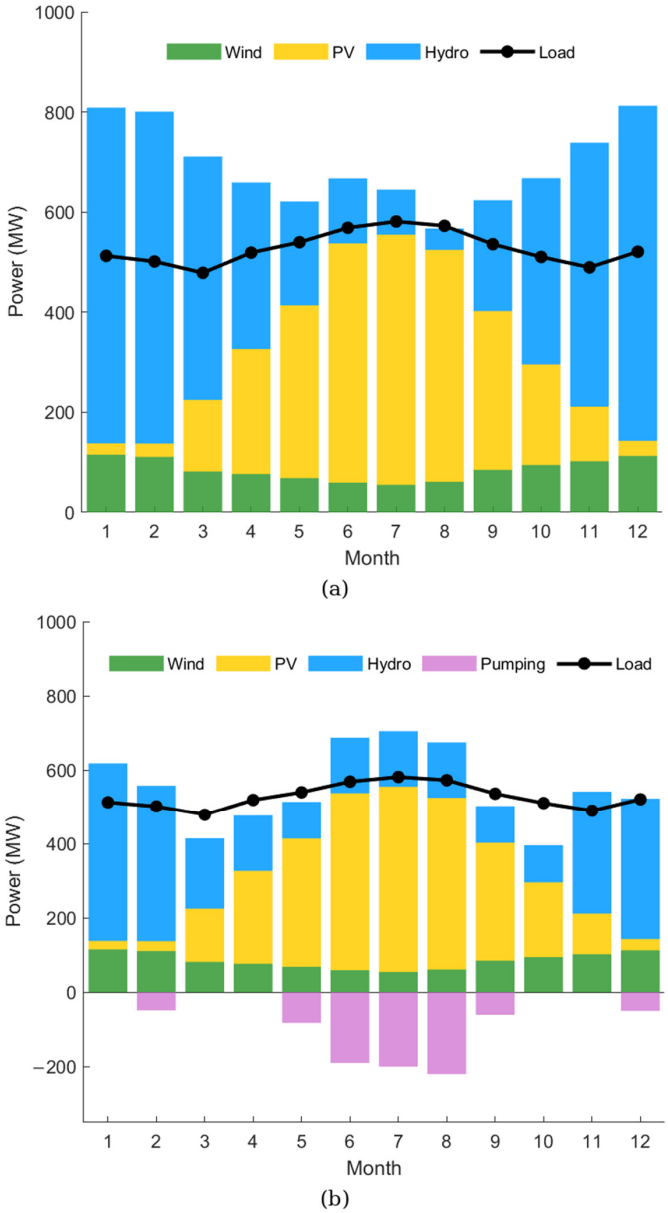


Figure 4. *Cont.*

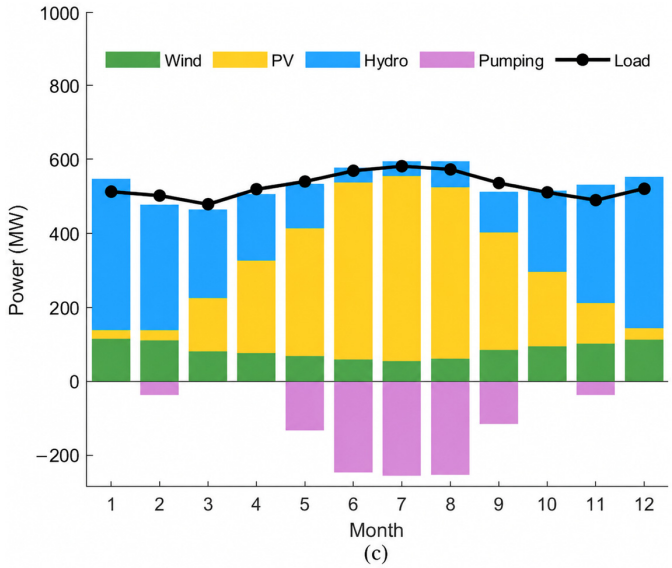


Figure 4. Output processes of the wind–PV hybrid system under the three schemes. (**a**) Scheme 1: no hybrid storage; (**b**) Scheme 2: hybrid daily regulation; and (**c**) Scheme 3: hybrid seasonal regulation.

The comparison of upper-reservoir water levels under the three schemes is shown in Figure 5. As Figure 5 shows, the water level in Scheme 1 exhibits sharp seasonal fluctuation, dropping rapidly to near the dead water level during the dry season and rebounding quickly during the flood season, reflecting the operating characteristic of a no-hybrid-storage system that lacks energy storage regulation means, with hydropower output entirely subject to the seasonal fluctuation of natural inflow and unable to actively smooth system output. After the hybrid storage function is introduced in Scheme 2, the amplitude of water-level fluctuation is markedly reduced, the dry-season water level is somewhat raised, and the drawdown during the transition season is gentler, reflecting the smoothing effect of hybrid storage regulation on short-term fluctuations, though no energy is transferred between seasons. Scheme 3 exhibits a pronounced reverse regulation pattern of "high in the flood season, low in the dry season": the dry-season water level falls to the lowest of the three schemes, while the flood-season water level rises to its highest and is sustained the longest, achieving cross-seasonal transfer of water resources by generating less and storing more during the flood season and generating more and releasing stored energy during the dry season.

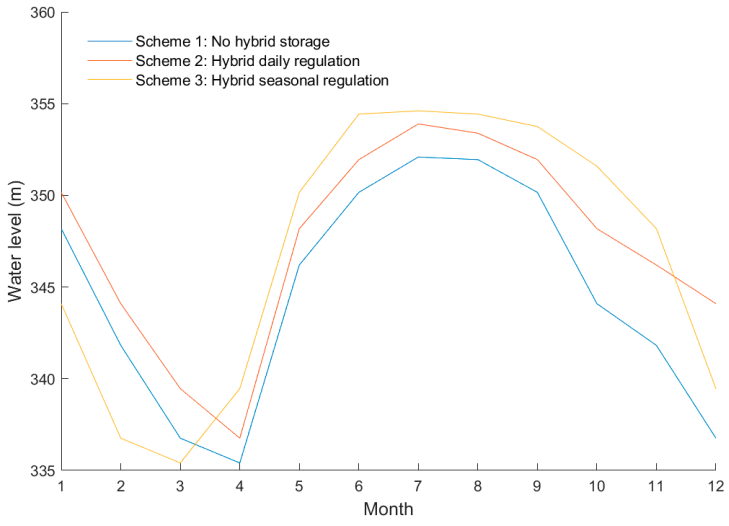


Figure 5. Comparison of upper-reservoir water levels under the three schemes.

3.3.2. Comprehensive Evaluation of Dispatch Performance Indicators

The comparison of key indicators under the three schemes is shown in Table 4, analyzed from the following three perspectives:

(1) Regarding renewable accommodation and output stability: Compared with the no-hybrid-storage scheme, the hybrid daily regulation scheme substantially reduces the amount of wind–PV curtailment through intraday pumping/generation balancing, significantly increasing the wind–PV accommodation rate and reducing the residual-load variance by about 69%. The hybrid seasonal regulation scheme further imposes cross-seasonal storage constraints, reducing wind–PV curtailment to a very low level and approaching full renewable accommodation, while reducing the residual-load variance by more than 77% relative to the no-hybrid-storage scheme.

(2) Regarding dry-season supply assurance and wet-season curtailment control: The no-hybrid-storage scheme has insufficient dry-season supply reliability and persistently high wet-season wind–PV curtailment. The hybrid daily regulation scheme improves dry-season supply reliability to 93.2%, but a residual seasonal supply–demand mismatch remains. The hybrid seasonal regulation scheme, through cross-seasonal water reallocation—storing in the flood season and releasing in the dry season—substantially improves dry-season supply reliability to 100% while reducing the wet-season wind–PV curtailment rate to 3.9%, effectively alleviating the resource complementarity conflict of "abundant water, weak wind–PV" versus "scarce water, strong wind–PV".

(3) Regarding overall operating benefit: The seasonal regulation scheme achieves a cross-seasonal energy transfer of 848.8 GWh, efficiently transferring surplus electricity from the wet season for use in the dry season, a capability the daily regulation scheme does not have. Although pumped storage incurs energy losses, seasonal regulation significantly improves water–energy coordination capability and system supply flexibility, confirming the important value of the hybrid station's seasonal regulation for the multi-energy complementary system.

Table 4. Comparison of key indicators under the three schemes.

Indicator	No Hybrid Storage	Hybrid, Daily Regulation	Hybrid, Seasonal Regulation
Wind–PV curtailment (GWh)	1682.26	335.93	82.74
Residual-load variance (MW^2)	12,589.28	3870.07	2861.05
Wind–PV accommodation rate	40.3%	87.4%	96.6%
Dry-season supply reliability	86.4%	93.2%	100%
Wet-season wind–PV curtailment rate	38.8%	13.4%	3.9%
Cross-seasonal transferred energy (GWh)	—	—	848.8

3.3.3. Seasonal Output Characteristics Analysis

The seasonal output characteristics of the wind–PV hybrid system under the three schemes are compared in Figure 6. Scheme 1 exhibits a pronounced seasonal mismatch between supply and demand: during the flood season, high hydropower output combined with wind–PV generation limits the available space for renewable energy accommodation, whereas during the dry season, reduced inflow constrains hydropower output and weakens the overall supply capability. Scheme 2 provides additional intraday regulation, but its storage and release remain confined mainly to the daily scale, which prevents surplus energy from being carried over across seasons and, therefore, limits its ability to alleviate the seasonal supply–demand mismatch. In contrast, Scheme 3 enables large-scale pumping during the flood season and releases the stored water during the dry season, thereby shifting energy across seasons and improving the seasonal balance between

renewable availability and system demand. These results demonstrate that the seasonal regulation strategy is essential for addressing the complementary mismatch between water resources and wind–PV output across seasons.

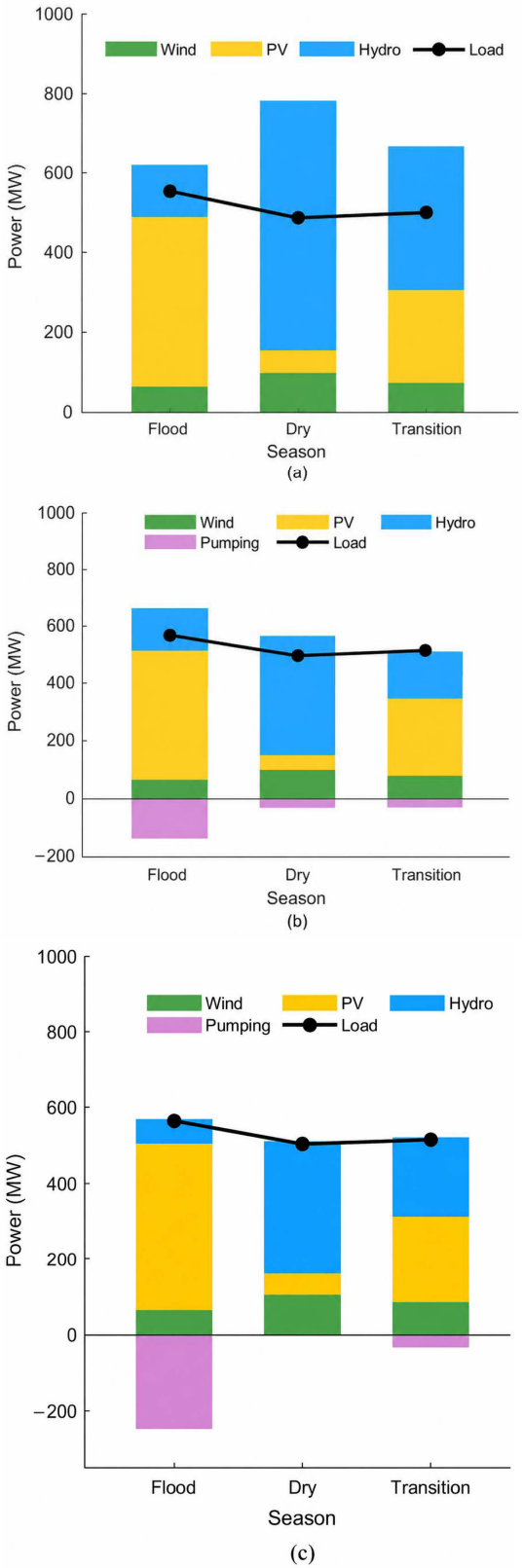


Figure 6. Seasonal output characteristics of the wind–PV hybrid system under the three schemes. (**a**) Scheme 1: no hybrid storage; (**b**) Scheme 2: hybrid daily regulation; and (**c**) Scheme 3: hybrid seasonal regulation.

3.3.4. Cross-Seasonal Energy Transfer Process

The cross-seasonal energy transfer process under the hybrid seasonal regulation scheme is shown in Figure 7. As Figure 7 shows, under the hybrid seasonal regulation scheme, the upper-reservoir water level and the cumulative net stored energy exhibit a pronounced reverse regulation pattern of storing water during the flood season and releasing it during the dry season. The dry-season water level is at its lowest level of the year, and the cumulative net stored energy also falls to its trough, indicating that the reservoir actively releases water to generate power, supplementing the dry-season power shortfall. Entering the flood season, as inflow increases and wind–PV output surges, the reservoir begins large-scale pumping and storage, with the water level continuously rising to around 355 m, while the cumulative net stored energy turns from negative to positive, peaking at 518 GWh and forming a distinct storage zone. During the transition seasons, the water level and stored energy change smoothly, achieving a smooth transition between the flood and dry seasons. The year-end water level is essentially level with the start of the year, and the cumulative net stored energy returns to near zero, demonstrating that the reservoir completes a full annual storage/release cycle and achieves a closed-loop cross-seasonal energy transfer. This process effectively confirms that the proposed seasonal regulation strategy can convert surplus electricity from the wet season into usable electricity for the dry season, improving the system's supply reliability.

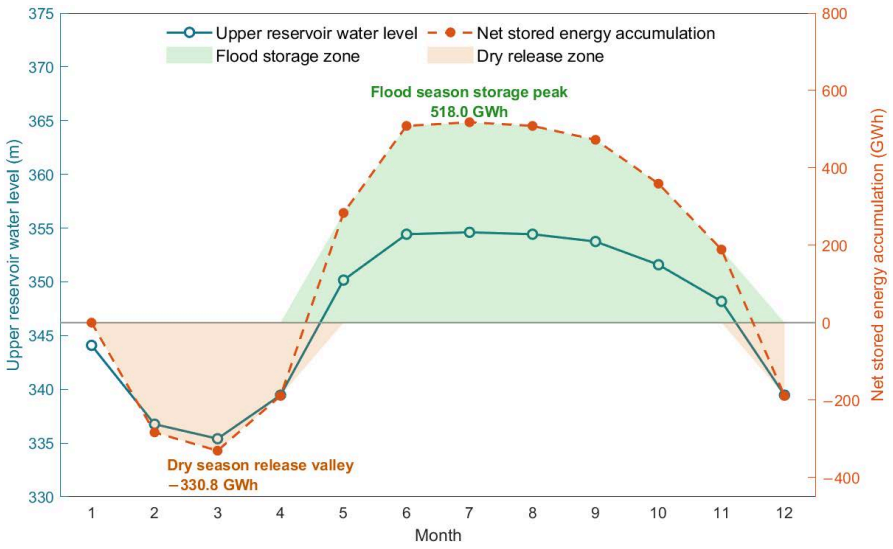


Figure 7. Cross-seasonal energy transfer process under the hybrid seasonal regulation scheme.

3.3.5. Typical-Day Dispatch Behavior Analysis

The comparison of typical-day output processes during the flood and dry seasons under the three schemes is shown in Figure 8. As Figure 8 shows, on the flood-season side in Scheme 1, PV output surges at midday, compounded by the high hydropower baseload, so the total output far exceeds the load, producing large-scale curtailment, and the system cannot accommodate the surplus renewable energy. Scheme 2 absorbs part of the curtailed power through midday pumping and supplements output with hybrid generation during the evening peak, markedly reducing curtailment, though the pumping period is concentrated and limited in scale. Scheme 3 further reduces the hydropower output to 620 MW, with the midday pumping power reaching as high as 450 MW, actively converting a large amount of curtailed power into reservoir-stored energy while keeping the total output closely matched to load, so curtailment nearly disappears, fully demonstrating the cross-seasonal storage behavior of "storing abundance" during the flood season. On the dry-season side, in Scheme 1, due to insufficient inflow, the hydropower output is

only 120 MW, and even with wind–PV added, there is still an all-day supply shortfall, giving extremely low reliability. Scheme 2 uses hybrid daily regulation to generate during the evening peak, narrowing the shortfall somewhat, though a considerable unmet load remains. Scheme 3, drawing on water energy stored during the flood season, raises the hydropower output to 350 MW, with hybrid generation sustained through the evening peak, so total output essentially covers load, and the shortfall (shown in red) is markedly reduced or eliminated, achieving cross-seasonal energy release that supplements the dry season with stored abundance. In summary, from flood-season curtailment accommodation to dry-season supply assurance, Scheme 3 outperforms both Scheme 2 and Scheme 1, intuitively confirming the comprehensive advantages of cross-seasonal optimal dispatch in smoothing fluctuations, improving accommodation, and enhancing reliability.

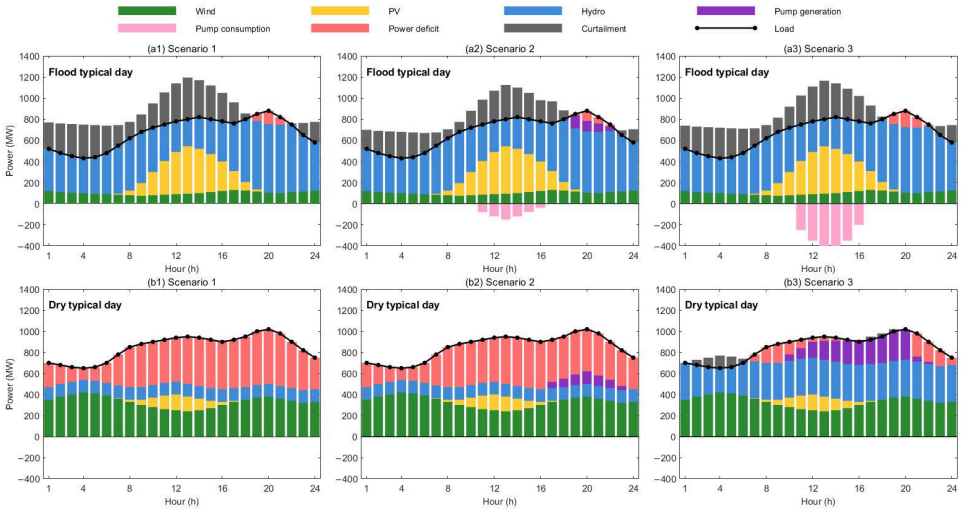


Figure 8. Comparison of typical-day output processes during the flood and dry seasons under the three schemes.

3.4. Multi-Year Validation and Uncertainty Scenario Analysis

To further examine the applicability and stability of the proposed cross-seasonal optimal dispatch method under different operating conditions, this section conducts analyses from two dimensions: variations in hydrological year types and operational uncertainties. First, wet, normal, and dry years are selected to compare the operating performance and key performance indicators of the three dispatch schemes under different inflow conditions, thereby further validating the multi-year applicability of the proposed method. On this basis, the typical uncertainty scenarios constructed in Section 2.2 are considered to further analyze the impacts of fluctuations in wind power, photovoltaic power, load, and reservoir inflow on system dispatch performance, and to quantitatively evaluate the dispatch adaptability of the proposed method under different uncertainty conditions.

3.4.1. Multi-Year Scenario Validation

The comparison of key performance indicators under different hydrological year types is presented in Table 5. As shown in Table 5, as the hydrological conditions shift from wet to normal and dry years, the residual-load variance generally increases for all three schemes, rising from 12,589.28 to 14,375.82 MW^2 for the no-hybrid-storage scheme, from 3870.07 to 5339.17 MW^2 for the hybrid daily regulation scheme, and from 2861.05 to 3904.50 MW^2 for the hybrid seasonal regulation scheme. Nevertheless, the hybrid seasonal regulation scheme consistently achieves the lowest residual-load variance across all three hydrological year types. Meanwhile, its renewable energy accommodation rate decreases from 96.6% in the wet year to 93.7% in the dry year, while the corresponding rates for the

no-hybrid-storage scheme decrease from 40.3% to 41.9% and those for the hybrid daily regulation scheme decrease from 87.4% to 75.4%. In terms of power supply reliability, the reliability of the no-hybrid-storage scheme decreases from 86.4% to 73.6%, that of the hybrid daily regulation scheme decreases from 93.2% to 86.1%, and that of the hybrid seasonal regulation scheme decreases from 100% to 97.3% from the wet to dry year. Overall, although deteriorating hydrological conditions impose greater regulation pressure on the system, the hybrid seasonal regulation scheme maintains the lowest residual-load variance, the highest renewable energy accommodation rate, and the highest power supply reliability across all three hydrological year types, demonstrating the applicability of the proposed method under different inflow conditions.

Table 5. Comparison of key performance indicators under different hydrological year types.

Hydrological Year Type	Dispatch Scheme	Residual-Load Variance (MW^2)	Wind–PV Accommodation Rate	Dry-Season Supply Reliability
Wet year	1	12,589.28	40.3%	86.4%
	2	3870.07	87.4%	93.2%
	3	2861.05	96.6%	100%
Normal year	1	13,890.46	41.1%	83.5%
	2	4585.37	83.6%	91.5%
	3	3084.74	95.2%	100%
Dry year	1	14,375.82	41.9%	73.6%
	2	5339.17	75.4%	86.1%
	3	3904.50	93.7%	97.3%

The comparison of residual-load variance under different hydrological year types is shown in Figure 9. As shown in Figure 9, the residual-load variance of all three schemes increases from wet to dry years, while the hybrid seasonal regulation scheme consistently maintains the lowest level. Notably, the increase in residual-load variance under the seasonal regulation scheme is relatively limited compared with the other two schemes, indicating that its load-smoothing advantage remains effective even under less favorable inflow conditions. This further demonstrates the stable performance of the proposed method across different hydrological year types.

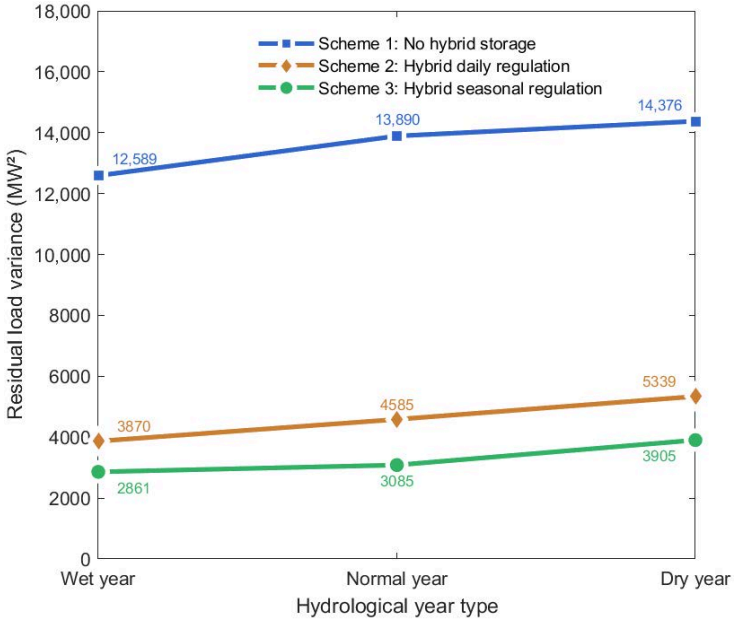


Figure 9. Comparison of residual-load variance under different hydrological year types.

3.4.2. Uncertainty Scenario Analysis

Based on the representative uncertainty scenarios constructed in Section 2.2, the proposed cross-seasonal optimal dispatch model is further evaluated under different operating conditions. The comparison of key performance indicators under different uncertainty scenarios is presented in Table 6. As shown in Table 6, different uncertainty conditions lead to distinct variations in system performance. The residual-load variance ranges from 2877.45 MW^2 to 9299.54 MW^2 among the six scenarios. The baseline scenario S1 achieves the lowest residual-load variance of 2877.45 MW^2, indicating strong load-smoothing capability under normal operating conditions, while the comprehensive adverse scenario S6 presents the highest variance of 9299.54 MW^2 due to the combined impacts of low wind–PV output, high load demand, and dry-year conditions. The wind power accommodation rate remains high across all scenarios, ranging from 96.9% to 100%, demonstrating the effectiveness of the proposed cross-seasonal regulation strategy in promoting renewable energy utilization. Specifically, full wind power accommodation is achieved in S2, S3, S5, and S6, while slight curtailment occurs in S1 and S4. For dry-season supply reliability, S1 and S4 maintain 100% reliability, whereas the reliability decreases under more challenging conditions, with S6 reaching 75.01%. These results indicate that although the system maintains excellent renewable accommodation capability, combined uncertainties in renewable generation, load demand, and hydrological conditions can significantly increase residual-load fluctuations and challenge dry-season supply security.

Table 6. Key performance indicators under different uncertainty scenarios.

Uncertainty Scenario	Residual-Load Variance (MW^2)	Wind–PV Accommodation Rate	Dry-Season Supply Reliability
S1	2877.45	96.9%	100%
S2	6265.17	100%	89.93%
S3	4918.19	100%	92.65%
S4	2907.29	97.4%	100%
S5	8232.53	100%	88.63%
S6	9299.54	100%	75.01%

To further evaluate the adaptability and robustness of the proposed cross-seasonal optimal dispatch method under complex uncertain conditions, the relative variations in key performance indicators compared with the baseline scenario S1 are analyzed, as shown in Figure 10. As illustrated in Figure 10a, different uncertainty scenarios result in varying degrees of increase in residual-load variance, indicating that fluctuations in renewable generation, load demand, and hydrological conditions introduce additional challenges to system operation. The residual-load variance increases by 117.7%, 70.9%, 1.0%, 186.1%, and 223.2% for S2–S6, respectively. Among these scenarios, S6 exhibits the largest increase due to the combined effects of low wind–PV output, high load demand, and dry-year conditions, demonstrating that multiple uncertainties can significantly intensify system operating fluctuations. Nevertheless, the variation characteristics under different scenarios also indicate that the proposed dispatch strategy can effectively adapt to diverse operating conditions.

Figure 10b presents the relative changes in dry-season supply reliability under different uncertainty scenarios. Compared with S1, the reliability decreases by 10.1%, 7.4%, 0%, 11.4%, and 25.0% for S2–S6, respectively. The most significant reduction occurs in S6, where multiple adverse factors are simultaneously considered. However, according to the results in Table 5, the wind power accommodation rate remains at 100% in S6, indicating that the system can still maintain strong renewable energy utilization capability under extreme uncertainty conditions. These results demonstrate that the proposed cross-seasonal

optimal dispatch method provides effective adaptability under different operating scenarios and enhances the operational resilience of the wind–PV hybrid storage multi-energy complementary system.

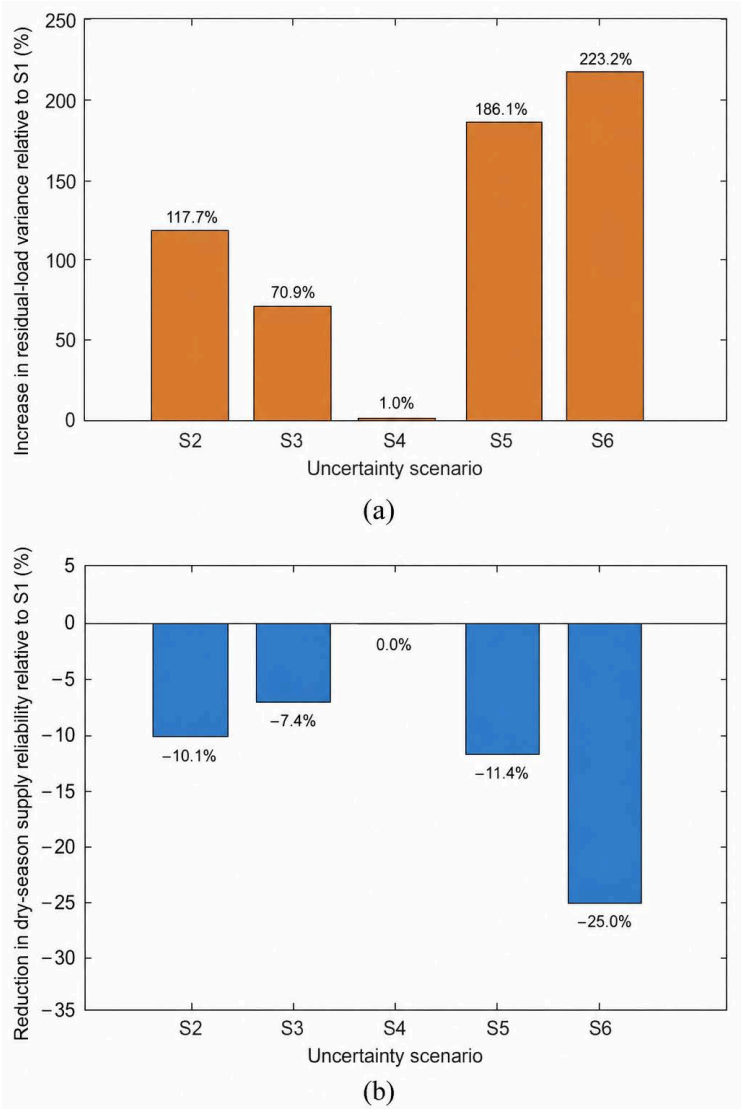


Figure 10. Impact of uncertainty scenarios on system performance. (**a**) Relative increase in residual-load variance under different uncertainty scenarios; (**b**) Relative decrease in dry-season supply reliability under different uncertainty scenarios.

3.4.3. Scalability Analysis

To further evaluate the scalability of VPPOA for larger-scale renewable energy systems, different renewable-energy-scale scenarios are considered. Three scale levels are investigated, corresponding to 1.0, 1.5, and 2.0 times the original renewable energy scale, respectively. The optimization performance under different scales is evaluated using residual-load variance, iteration number, and computational time, as summarized in Table 7. As shown in Table 7, increasing the renewable energy scale leads to a gradual increase in residual-load variance, iteration number, and computational time due to the increased complexity of the optimization problem. Nevertheless, VPPOA maintains stable convergence and satisfactory optimization performance even when the renewable energy scale is doubled. The moderate increase in computational cost demonstrates that the proposed VPPOA has good scalability and can effectively solve larger-scale optimization problems.

Table 7. Scalability analysis of VPPOA under different renewable energy scales.

Renewable Energy Scale	Residual-Load Variance (MW^2)	Iteration Number	Computational Time (s)
1×	2877.45	117	34.63
1.5×	2958.42	143	42.18
2×	3108.76	168	55.64

3.5. Performance Evaluation of the Proposed VPPOA

To verify the effectiveness of the proposed VPPOA for the cross-seasonal optimal dispatch model, VPPOA is compared with GA, PSO, and POA. The performance of the proposed algorithm is evaluated from the perspectives of convergence performance, stability, and comprehensive optimization performance. Furthermore, the influence of key parameter variations on the algorithm performance is investigated. All simulations are performed in MATLAB R2023b on a computer equipped with an Intel Core i5-1155G7 processor and 16 GB of RAM. To ensure a fair comparison, all algorithms are tested under the same dispatch model, constraints, and computational environment. The parameters of VPPOA are set as follows: the maximum iteration number is 200, the convergence tolerance is 1×10^{-4}, and the stage sequence is set to [1,2,3,2,1]. These parameters are determined based on preliminary tests, achieving a balance between solution accuracy and computational efficiency, and providing the basis for subsequent algorithm performance comparison and parameter sensitivity analysis.

3.5.1. Comparison of Optimization Algorithm Performance

The comparison of convergence performance among different optimization algorithms is shown in Figure 11. As shown in Figure 11, the residual-load variance of all algorithms decreases with the increase in iterations, indicating that all algorithms have certain optimization capabilities. Compared with GA, PSO, and POA, VPPOA achieves faster convergence and obtains the lowest final objective value. This demonstrates that the proposed stage-based search strategy improves the optimization efficiency and convergence performance of VPPOA.

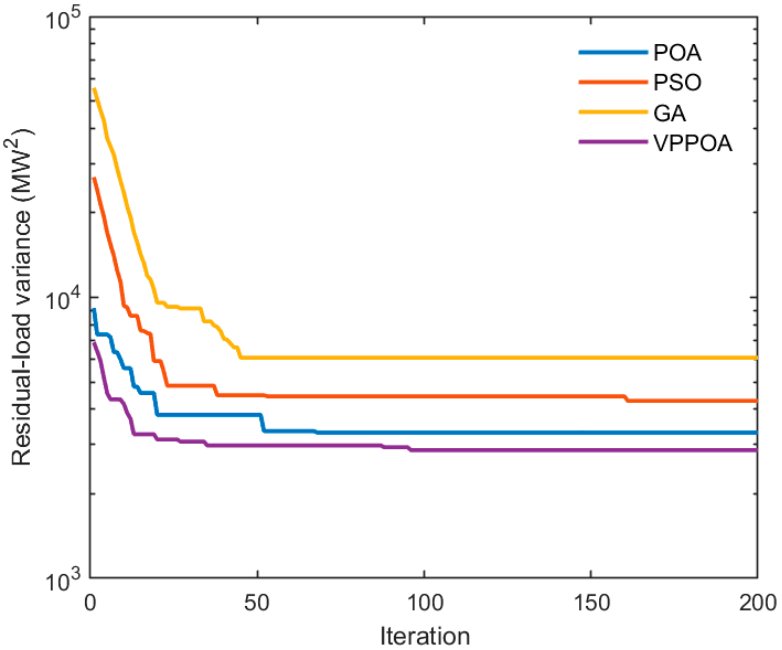


Figure 11. Convergence curves of different optimization algorithms.

The stability comparison of different optimization algorithms is presented in Figure 12. As shown in Figure 12, the boxplots are obtained from 30 independent runs of each algorithm. VPPOA achieves the lowest median objective value and the smallest distribution range among all algorithms, indicating that it can obtain high-quality solutions with lower fluctuations. The smaller dispersion of VPPOA demonstrates its lower sensitivity to random initialization and stronger robustness.

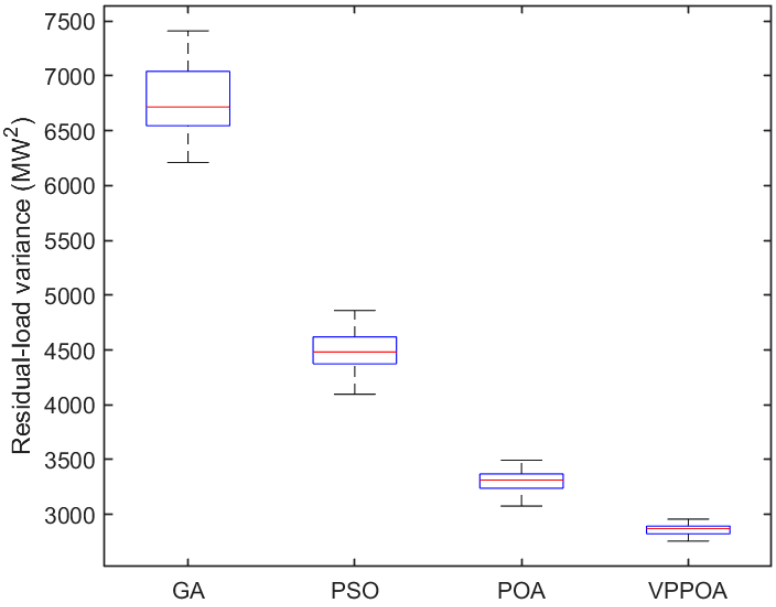


Figure 12. Stability comparison of different optimization algorithms.The blue boxes represent the interquartile ranges, the red lines indicate the medians, and the black lines represent the whiskers.

The comprehensive performance comparison of different optimization algorithms is presented in Table 8. As shown in Table 8, VPPOA achieves the lowest residual-load variance, reaching 2861.05 MW^2, which is reduced by 56.3%, 41.3%, and 20.1% compared with GA, PSO, and POA, respectively. Meanwhile, VPPOA obtains the highest wind–PV accommodation rate of 96.6%, demonstrating its superior optimization capability. In terms of computational efficiency, VPPOA requires 34.63 s, which is slightly higher than POA with 32.95 s. This is because the introduced stage-based search mechanism increases the computational burden to some extent. However, it can reduce ineffective exploration and improve the optimization capability. Overall, VPPOA achieves a better balance between optimization performance and computational efficiency.

Table 8. Comprehensive performance comparison of different optimization algorithms.

Algorithm	Residual-Load Variance (MW^2)	Wind–PV Accommodation Rate	Computational Time (s)
GA	6551.13	78.3%	42.61
PSO	4876.28	84.2%	39.48
POA	3580.77	95.2%	32.95
VPPOA	2861.05	96.6%	34.63

3.5.2. Parameter Sensitivity Analysis

The influence of stage sequence on the performance of VPPOA is presented in Table 9. As shown in Table 9, the residual-load variance obtained under different stage sequences remains within a relatively small range, indicating that VPPOA has good adaptability to the stage sequence parameter. Among all cases, the stage sequence [1,2,3,2,1] achieves

the lowest residual-load variance of 2861.05 MW^2 and requires only 117 iterations for convergence, demonstrating better optimization performance and search efficiency. In contrast, excessive or insufficient stage variations may slightly reduce the optimization performance, indicating that an appropriate stage sequence can effectively balance exploration and exploitation. Therefore, the selected stage sequence [1,2,3,2,1] is reasonable for the proposed VPPOA.

Table 9. Sensitivity analysis of different stage sequences.

Algorithm	Residual-Load Variance (MW^2)	Iteration
[1 1 1 1 1]	2864.48	126
[1 2 2 2 1]	2863.32	122
[1 2 3 2 1]	2861.05	117
[1 3 5 3 1]	2867.47	133
[2 3 4 3 2]	2867.87	138

4. Discussion

The results demonstrate that introducing seasonal regulation capability into hybrid pumped-storage stations can significantly enhance the flexibility of wind–PV–hydro complementary systems. Unlike conventional daily regulation strategies that mainly address short-term renewable fluctuations, seasonal regulation operation enables coordinated water storage and energy release across different seasons. Thus, renewable energy surplus during wet periods can be converted into regulating resources for dry periods, which explains the improvement in renewable energy accommodation and reduction in residual-load fluctuations.

The annual cross-seasonal energy transfer of 848.8 GWh further reveals the mechanism of the proposed strategy. By introducing seasonal reservoir storage rate-of-change constraints, the model establishes a long-term storage trajectory instead of restoring the storage state within each daily cycle. This effectively mitigates the mismatch between renewable generation and seasonal hydrological variations. Compared with hybrid daily regulation, the seasonal regulation strategy improves dry-season supply reliability from 93.2% to 100%, demonstrating the importance of cross-seasonal storage for power supply security under unfavorable hydrological conditions.

Beyond improving renewable accommodation, the seasonal regulation capability of hybrid pumped-storage stations also provides long-duration flexibility for future renewable-dominated power systems. By coordinating wet-season storage and dry-season release, the proposed strategy improves renewable utilization and enhances the operational resilience of hydro–wind–PV complementary systems. Therefore, exploiting the long-term storage potential of hybrid pumped-storage stations is of significant engineering value for renewable energy bases with strong seasonal hydrological characteristics.

The multi-scenario analysis verifies the robustness of the proposed framework under different hydrological conditions, renewable output levels, and load variations. Meanwhile, comparisons with GA, PSO, and traditional POA demonstrate the effectiveness of VPPOA for complex multi-time-scale scheduling problems. However, the proposed model mainly focuses on medium-to-long-term cross-seasonal allocation and does not explicitly model detailed intraday pumping and generation scheduling. Although representative hydrological scenarios are considered, probabilistic uncertainty of future inflow and renewable generation is not explicitly modeled. Future research could integrate nested long-short-term scheduling and stochastic or robust optimization methods to further improve practical applicability.

5. Conclusions

This paper investigates the cross-seasonal optimal dispatch of a wind–PV hybrid-storage multi-energy complementary system considering the seasonal regulation characteristics of hybrid pumped-storage stations. A medium-to-long-term optimal dispatch model with seasonal reservoir storage rate-of-change constraints is established, and the Variable-Phase Progressive Optimality Algorithm (VPPOA) is developed to solve the proposed model. The effectiveness of the proposed framework is validated through comparisons among different operation strategies, multiple scenarios, and optimization algorithms.

The results show that the seasonal regulation strategy can effectively exploit the long-term storage potential of hybrid pumped-storage stations. Compared with the no-hybrid-storage scheme, the hybrid seasonal regulation strategy reduces residual-load variance by 77.3%, increases the wind–PV accommodation rate from 40.3% to 96.6%, and achieves 848.8 GWh of annual cross-seasonal energy transfer. Compared with the hybrid daily regulation strategy, it further improves dry-season supply reliability from 93.2% to 100%.

The multi-scenario validation and algorithm comparison further demonstrate the robustness and effectiveness of the proposed framework. The results highlight the importance of considering cross-seasonal regulation capability in hybrid pumped-storage operation and provide an effective approach for enhancing the flexibility and reliability of future renewable-dominated power systems.

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Abbreviations

The following abbreviations are used in this manuscript:

PV	Photovoltaic
VPPOA	Variable-Phase Progressive Optimality Algorithm
MILP	Mixed-Integer Linear Programming
ECMWF	European Centre for Medium-Range Weather Forecasts

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